

PD Control of Robot Manipulators with Neural Compensation

Lyapunov design · stability analysis · learning laws

Wen Yu

Departamento de Control Automático

CINVESTAV-IPN

A.P. 14-740, Av. IPN 2508, Ciudad de México, 07360, México

Part I – Model and classical PD

- ➊ Rigid-robot dynamics and structural properties
- ➋ Passivity: the meaning of skew-symmetry
- ➌ PD regulation: Lyapunov + LaSalle

Part II – Neural compensation: *regulation*

- ➍ Neural approximation of unknown functions
- ➎ Setting: the unknown is only $G(q) + F(\dot{q})$
- ➏ Why the natural V_4 is not enough
- ➐ Cross term $\alpha e^T M \dot{q}$; birth of $r = \dot{e} + \alpha e$
- ➑ Bias $\propto \bar{e}$, not $\propto k_g + k_f$; design recipe

Part III – Tracking with a known model

- ➒ The filtered error $r = \dot{e} + \Lambda e$, general Λ
- ➓ Error dynamics $M\dot{r} = -Cr + \tau - h(x)$
- ➔ Exponential stability; PD alone only UUB

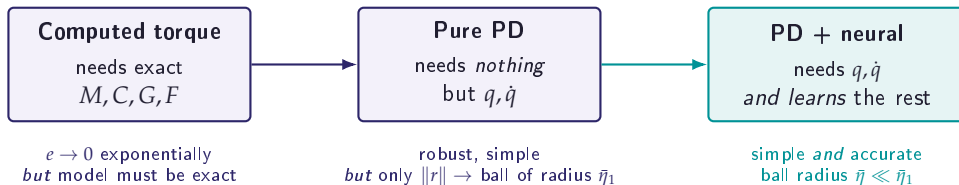
Part IV – Neural compensation: *tracking*

- ➔ Architecture $\tau = -Kr + \hat{W}_t \sigma(x)$
- ➕ Lyapunov analysis and the learning law
- ➖ Main theorem: a smaller residual set; robustness

Central message

Learning does not replace the PD loop — it **shrinks the residual set** PD alone guarantees.

Motivation: three levels of knowledge



Why this matters in practice

- ▶ Industrial manipulators ship with PD/PID loops: we want to **keep the existing loop** and add a compensator.
- ▶ $G(q)$ and $F(\dot{q})$ (Coulomb friction, backlash, payload changes) are the dominant unmodelled terms.
- ▶ The neural block is a *feedforward* term; the PD term still provides the stabilising damping.

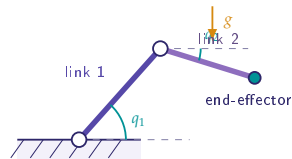
Dynamics of a serial n -link rigid manipulator

$$M(q) \ddot{q} + C(q, \dot{q}) \dot{q} + G(q) + F(\dot{q}) = \tau$$

$q \in \mathbb{R}^n$	joint positions
$M(q) \in \mathbb{R}^{n \times n}$	inertia matrix
$C(q, \dot{q}) \dot{q}$	Coriolis / centrifugal
$G(q) = \partial P(q) / \partial q$	gravity
$F(\dot{q}) = F_v \dot{q} + F_c \operatorname{sgn}(\dot{q})$	friction
$\tau \in \mathbb{R}^n$	applied torque

Obtained from the Euler-Lagrange equation

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \tau, \text{ with } L = \frac{1}{2} \dot{q}^T M(q) \dot{q} - P(q).$$



Structural properties (the whole analysis rests on these)

P1 — Inertia matrix

$M(q) = M(q)^T > 0$ and uniformly bounded:

$$m_1 \|x\|^2 \leq x^T M(q)x \leq m_2 \|x\|^2, \quad \forall x \in \mathbb{R}^n$$

$\Rightarrow$ any $V = \frac{1}{2} \dot{r}^T M r$ is a legitimate (decreascent, radially unbounded in r) Lyapunov function.

P2 — Skew-symmetry

$$x^T [\dot{M}(q) - 2C(q, \dot{q})]x = 0, \quad \forall x \in \mathbb{R}^n$$

i.e. $\dot{M} - 2C$ is skew-symmetric.

P3 — Boundedness of C

$$\|C(q, \dot{q})x\| \leq k_c \|\dot{q}\| \|x\|$$

C is linear in $\dot{q}$: the quadratic growth is what forces the “ $\ll$ ” bounds later.

P4 — Gravity and friction

$$\|G(q)\| \leq k_g, \quad \dot{q}^T F(\dot{q}) \geq 0$$

Friction is *dissipative* but non-smooth (sgn)
 $\Rightarrow$ hard to model, easy to learn.

P1 builds the Lyapunov function • P2 cancels the Coriolis term • P3–P4 bound the disturbance

Why skew-symmetry? Energy and passivity

Total energy $\mathcal{E} = \underbrace{\frac{1}{2}\dot{q}^T M(q)\dot{q}}_{\text{kinetic}} + \underbrace{P(q)}_{\text{potential}}$, with $G(q) = \partial P / \partial q$.

$$\begin{aligned}\dot{\mathcal{E}} &= \dot{q}^T M \ddot{q} + \frac{1}{2} \dot{q}^T \dot{M} \dot{q} + \dot{q}^T G(q) \\ &= \dot{q}^T [\tau - C\dot{q} - G - F] + \frac{1}{2} \dot{q}^T \dot{M} \dot{q} + \dot{q}^T G \\ &= \dot{q}^T \tau - \dot{q}^T F(\dot{q}) + \underbrace{\frac{1}{2} \dot{q}^T [\dot{M} - 2C] \dot{q}}_{=0 \text{ by P2}} = \dot{q}^T \tau - \dot{q}^T F(\dot{q}) \leq \dot{q}^T \tau\end{aligned}$$

Interpretation

The map $\tau \mapsto \dot{q}$ is **passive**: internal (Coriolis) forces do no net work, they only *redistribute* kinetic energy between joints.

Consequence for design

Every $\frac{1}{2} \dot{x}^T \dot{M} x - x^T C x$ that appears below *vanishes*. This is why a simple PD term suffices — no cancellation of C is needed.

Regulation: control objective and closed loop

Objective. q^d constant ($\dot{q}^d = \ddot{q}^d = 0$); drive $e := q - q^d \rightarrow 0$. Note $\dot{e} = \dot{q}$.

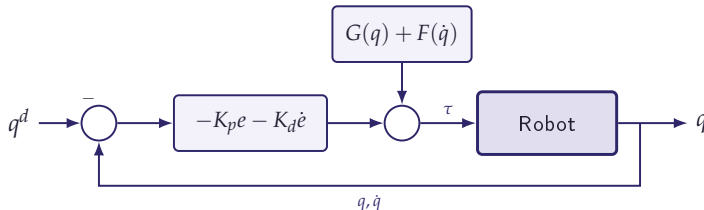
PD with gravity/friction compensation

$$\tau = -K_p(q - q^d) - K_d(\dot{q} - \dot{q}^d) + G(q) + F(\dot{q}) = -K_p e - K_d \dot{e} + G(q) + F(\dot{q})$$

with $K_p = K_p^T > 0$, $K_d = K_d^T > 0$ constant.

Substituting into $M\ddot{q} = \tau - C\dot{q} - G - F$ gives the **closed loop**

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + K_p e + K_d \dot{e} = 0$$



Regulation: Lyapunov analysis, step by step

Candidate: total closed-loop energy

$$V_1 = \frac{1}{2} \dot{e}^T M(q) \dot{e} + \frac{1}{2} e^T K_p e \geq \frac{1}{2} m_1 \|\dot{e}\|^2 + \frac{1}{2} \lambda_{\min}(K_p) \|e\|^2 > 0$$

(positive definite and radially unbounded by **P1**).

$$\dot{V}_1 = \dot{e}^T M \ddot{e} + \frac{1}{2} \dot{e}^T \dot{M} \dot{e} + e^T K_p \dot{e} \quad (\text{chain rule, } \ddot{e} = \ddot{q})$$

$$= \dot{q}^T [-C\dot{q} - K_p e - K_d \dot{q}] + \frac{1}{2} \dot{q}^T \dot{M} \dot{q} + \dot{q}^T K_p e \quad (\text{closed loop, } \dot{e} = \dot{q})$$

$$= \underbrace{\frac{1}{2} \dot{q}^T [\dot{M} - 2C] \dot{q}}_{=0 \text{ (P2)}} \underbrace{- \dot{q}^T K_p e + \dot{q}^T K_p e}_{=0 \text{ (cancel)}} - \dot{q}^T K_d \dot{q}$$

Result

$$\dot{V}_1 = -\dot{q}^T K_d \dot{q} \leq -\lambda_{\min}(K_d) \|\dot{q}\|^2 \leq 0$$

$\Rightarrow$ the equilibrium $(e, \dot{e}) = (0, 0)$ is *stable*, and $e, \dot{q} \in \mathcal{L}_\infty, \dot{q} \in \mathcal{L}_2$.

Regulation: from stability to asymptotic stability

$\dot{V}_1 \leq 0$ is only *negative semi-definite* ($\dot{V}_1 = 0$ whenever $\dot{q} = 0$, for *any* e).

LaSalle's invariance principle

Let $S = \{(e, \dot{e}) : \dot{V}_1 = 0\} = \{(e, \dot{e}) : \dot{q} = 0\}$. On any trajectory that stays in S :

$$\dot{q} \equiv 0 \Rightarrow \ddot{q} \equiv 0 \xrightarrow{\text{closed loop}} K_p e \equiv 0 \xrightarrow{K_p > 0} e \equiv 0$$

The largest invariant set in S is the origin $\Rightarrow$ **global asymptotic stability**.

Remark 1 — PD without gravity

With $\tau = -K_p e - K_d \dot{e}$ only, the steady state satisfies $K_p e = -G(q)$, hence a *bias* $\|e_\infty\| \leq k_g / \lambda_{\min}(K_p)$. Large K_p reduces it but amplifies noise.

Remark 2 — why go further

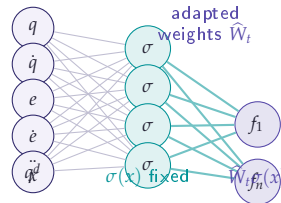
Regulation is easy because $\dot{q}^d = 0$ removes all the inertial and Coriolis terms. For **tracking** ($\dot{q}^d \neq 0$) the same PD no longer cancels anything — this is the real problem.

Neural approximation of unknown functions

Let $f(x)$ be any continuous unknown function of a measurable x . On a compact set $\mathcal{X} \ni x$, universal approximation gives

$$f(x) = W^* \sigma(x) + \epsilon(x), \quad \|\epsilon(x)\| \leq \bar{\epsilon}$$

- ▶ $\sigma(x) \in \mathbb{R}^m$: fixed basis (sigmoids with random inner weights, RBFs, wavelets);
- ▶ $W^* \in \mathbb{R}^{n \times m}$: unknown “ideal” weights, $W^* = \arg \min_W \sup_{x \in \mathcal{X}} \|f(x) - W\sigma(x)\|$;
- ▶ $\epsilon(x)$: approximation error, $\bar{\epsilon} \downarrow$ as $m \uparrow$.



Key structural choice — used twice

Only the **output layer is adapted** $\Rightarrow$ linear in the parameters $\Rightarrow$ a Lyapunov learning law exists, with no backpropagation and no local minima. We apply it to $f = G + F$ (regulation, $x \in \mathbb{R}^{2n}$) and to $f = h(x)$ (tracking, $x \in \mathbb{R}^{7n}$).

Regulation with neural compensation: setting

Now q^d is *constant*: $e = q - q^d$, $\dot{e} = \dot{q}$. All inertial and Coriolis feedforward terms disappear and the **only** unknown is

$$g(x) := G(q) + F(\dot{q}), \quad x = [q^T, \dot{q}^T]^T \in \mathbb{R}^{2n}.$$

Neural approximation

$$g(x) = W^* \sigma(x) + \epsilon(x), \quad \|\epsilon(x)\| \leq \bar{\epsilon}$$

PD + neural compensator (regulation)

$$\tau = -K_p e - K_d \dot{q} + \hat{W}_t \sigma(x)$$

The baseline to beat

Pure PD ($\hat{W}_t \equiv 0$) leaves a *static bias*: at equilibrium $K_p e_\infty = -g$, so

$$\|e_\infty\| \leq \frac{k_g + k_f}{\lambda_{\min}(K_p)}.$$

Why regulation is the easy case

- ▶ input dimension $2n$ instead of $7n$: far fewer basis functions;
- ▶ at rest $\dot{q} \rightarrow 0$, so $\hat{W}\sigma(q, 0)$ is a **learned gravity map** $\hat{G}(q)$ — storable and reusable.

Regulation: closed loop and a first (insufficient) Lyapunov function

With $g = W^* \sigma + \epsilon$ and $\tilde{W}_t = \hat{W}_t - W^*$ the closed loop is

$$M(q) \ddot{q} + C(q, \dot{q}) \dot{q} + K_p e + K_d \dot{q} = \tilde{W}_t \sigma(x) - \epsilon.$$

Candidate V_4

$$V_4 = \frac{1}{2} \dot{q}^T M \dot{q} + \frac{1}{2} e^T K_p e + \frac{1}{2} \text{tr}(\tilde{W}_t^T K_w^{-1} \tilde{W}_t)$$

$$\dot{V}_4 = \underbrace{\frac{1}{2} \dot{q}^T [\dot{M} - 2C] \dot{q}}_{=0 \text{ (P2)}} - \dot{q}^T K_d \dot{q} - \dot{q}^T \epsilon$$

$$+ \text{tr}[\tilde{W}_t^T (K_w^{-1} \hat{W}_t + \dot{q} \sigma^T)]$$

With $\hat{W}_t = -K_w \dot{q} \sigma^T(x)$: $\dot{V}_4 \leq -\dot{q}^T (K_d - \Lambda) \dot{q} + \bar{\eta}$.

Two defects

- 1 $\dot{V}_4$ is negative *semi*-definite: it bounds $\dot{q}$ but says **nothing about e** ; LaSalle is unavailable because $\bar{\eta} \neq 0$.
- 2 The update is driven by $\dot{q}$ alone: once the arm settles, $\dot{q} \rightarrow 0$ and **learning stops** — exactly where the bias lives.

Remedy

Add a **cross term** $\alpha e^T M(q) \dot{q}$ to V_4 : it injects $-\alpha e^T K_p e$ into $\dot{V}$ and puts e into the learning law.

Regulation: cross-term Lyapunov function and learning law

Candidate V_5

$$V_5 = \frac{1}{2} \dot{q}^T M \dot{q} + \frac{1}{2} e^T K_p e + \alpha e^T M(q) \dot{q} + \frac{1}{2} \text{tr}(\tilde{W}_t^T K_w^{-1} \tilde{W}_t), \quad \alpha > 0$$

$$V_5 > 0 \iff \begin{bmatrix} K_p & \alpha M \\ \alpha M & M \end{bmatrix} > 0 \iff K_p > \alpha^2 M, \text{ guaranteed by } \alpha < \sqrt{\lambda_{\min}(K_p)/m_2}.$$

Using the Christoffel form of C ($\mathbf{P2}'$: $\dot{M} = C + C^T$), so that $e^T \dot{M} \dot{q} - e^T C \dot{q} = \dot{q}^T C e$:

$$\frac{d}{dt}(\alpha e^T M \dot{q}) = \alpha \left[\dot{q}^T M \dot{q} + \dot{q}^T C e - e^T K_p e - e^T K_d \dot{q} + e^T (\tilde{W}_t \sigma - \epsilon) \right]$$

Modified learning law — a *filtered error* is born

The trace terms now cancel with

$$\hat{W}_t = -K_w (\dot{q} + \alpha e) \sigma^T(x) = -K_w r \sigma^T(x), \quad r := \dot{e} + \alpha e$$

The cross term has manufactured the signal $r = \dot{e} + \alpha e$: learning no longer stops when $\dot{q} = 0$. In Part III this same r — with a general matrix Λ in place of αI — becomes the central object of the tracking design.

Regulation: main theorem and the size of the residual bias

Collecting terms, with **P1**, **P3** on $\|e\| \leq \rho_e$:

$$\dot{V}_5 \leq -\gamma_v \|\dot{q}\|^2 - \gamma_p \|e\|^2 + \bar{\epsilon} (\|\dot{q}\| + \alpha \|e\|)$$

with $\gamma_v = \lambda_{\min}(K_d) - \alpha \left[m_2 + k_c \rho_e + \frac{\lambda_{\max}^2(K_d)}{2\lambda_{\min}(K_p)} \right],$

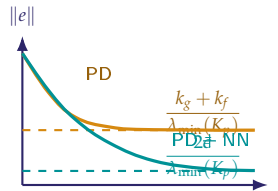
$$\gamma_p = \frac{\alpha}{2} \lambda_{\min}(K_p).$$

Theorem (PD + neural compensation, regulation)

If $\alpha < \sqrt{\lambda_{\min}(K_p)/m_2}$ and $\gamma_v > 0$, then $e, \dot{q}, \hat{W}_t$ are bounded and

$$\limsup_{t \rightarrow \infty} \|e\| \leq \frac{2\bar{\epsilon}}{\lambda_{\min}(K_p)}, \quad \limsup_{t \rightarrow \infty} \|\dot{q}\| \leq \frac{\bar{\epsilon}}{\gamma_v}.$$

Semiglobal: enlarging ρ_e is paid for with a larger K_d .



Same K_p, K_d : the bias is no longer set by the full gravity/friction torque, but by the approximation error $\bar{\epsilon}$.

Special case: linear in the parameters

If $g(q) = Y(q)\theta^*$ exactly, then $\bar{e} = 0$ and $\hat{\theta} = -K_w Y^T(q)r \Rightarrow e \rightarrow 0$ asymptotically: classical *adaptive PD with gravity compensation* (Tomei 1991; Kelly 1997). The network is its unstructured extension — it also absorbs Coulomb friction, backlash and payload changes, for which no regressor exists.

Robustness

Since $\bar{e} \neq 0$, add e -modification or a dead-zone on $\|r\|$; projection keeps $\|\hat{W}_t\|_F \leq W_{\max}$, which bounds the torque.

Design recipe

- 1 Tune K_p, K_d as a stable PD loop first ($\hat{W}_0 = 0$).
- 2 Pick $\alpha \approx 0.3\sqrt{\lambda_{\min}(K_p)/m_2}$; check $\gamma_v > 0$.
- 3 RBF centres on a grid over the reachable q ; coarse $\dot{q}$ -channel.
- 4 Small K_w first, increase until the bias falls without oscillation.
- 5 Freeze $\hat{W}$ and store $\hat{G}(q) = \hat{W}\sigma(q, 0)$ as a table.

Preview: what changes for tracking (Parts III–IV)

	Regulation	Tracking
unknown	$G + F$	$M\ddot{q}^r + C\dot{q}^r + G + F$
input x	$[q, \dot{q}]$	$[q, \dot{q}, e, \dot{e}, q^d, \dot{q}^d, \ddot{q}^d]$
r	$\dot{e} + \alpha e$	$\dot{e} + \Lambda e$
law	$\hat{W}_t = -K_w r \sigma^T(x)$	

Tracking: the filtered error $r = \dot{e} + \Lambda e$

Now $q^d = q^d(t)$: $e = q - q^d(t)$, $\dot{e} = \dot{q} - \dot{q}^d(t)$. Keep the signal produced by the regulation cross term, with a general gain matrix ($\Lambda = \alpha I$ recovers Part II):

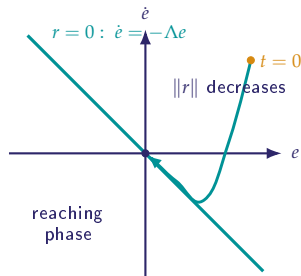
$$r = \dot{e} + \Lambda e, \quad \Lambda = \Lambda^T > 0$$

Why one signal instead of two

r is the output of the *stable* filter $e(s) = (sI + \Lambda)^{-1}r(s)$, so

- ▶ $r \in \mathcal{L}_\infty \Rightarrow e, \dot{e} \in \mathcal{L}_\infty$;
- ▶ $r \rightarrow 0 \Rightarrow e \rightarrow 0$ exponentially with rate $\lambda_{\min}(\Lambda)$;
- ▶ $\|r\| \leq \varepsilon \Rightarrow \limsup \|e\| \leq \varepsilon / \lambda_{\min}(\Lambda)$.

The n -dof second-order problem becomes a **first-order problem** in r .



The controller first drives the state onto the line $r = 0$; the line itself carries $e \rightarrow 0$.

Tracking: the error dynamics in terms of r

Define the **reference velocity/acceleration**

$$\dot{q}^r := \dot{q}^d - \Lambda e, \quad \ddot{q}^r = \ddot{q}^d - \Lambda \dot{e} \quad \implies \quad r = \dot{q} - \dot{q}^r.$$

Then, using $M\ddot{q} = \tau - C\dot{q} - G - F$ and $\dot{q} = r + \dot{q}^r$:

$$\begin{aligned} M\dot{r} &= M\ddot{q} - M\ddot{q}^r = \tau - C(r + \dot{q}^r) - G - F - M\ddot{q}^r \\ &= -Cr + \tau - \underbrace{\left[M(q)\ddot{q}^r + C(q, \dot{q})\dot{q}^r + G(q) + F(\dot{q}) \right]}_{=: h(x)} \end{aligned}$$

Compact error model

$$\boxed{M(q)\dot{r} = -C(q, \dot{q})r + \tau - h(x)} \quad x = [q^T, \dot{q}^T, e^T, \dot{e}^T, q^{dT}, \dot{q}^{dT}, \ddot{q}^{dT}]^T$$

$h(x)$ collects *all* the model information: inertia, Coriolis, gravity, friction. It is a **static nonlinear function** of the measurable x , so Part II's network applies verbatim: $h(x) = W^*\sigma(x) + \epsilon(x)$ — only the input dimension grows from $2n$ to $7n$.

Model-based tracking: exponential stability

If $h(x)$ is *known* choose

$$\tau = -Kr + h(x), \quad K = K^T > 0 \quad \implies \quad M\dot{r} = -Kr - Cr.$$

Lyapunov function $V_2 = \frac{1}{2}r^T M(q)r$

$$\begin{aligned}\dot{V}_2 &= r^T M\dot{r} + \frac{1}{2}r^T \dot{M}r = -r^T Kr - r^T Cr + \frac{1}{2}r^T \dot{M}r \\ &= -r^T Kr + \underbrace{\frac{1}{2}r^T [\dot{M} - 2C]r}_{=0 \text{ (P2)}} = -r^T Kr \leq -\lambda_{\min}(K) \|r\|^2\end{aligned}$$

With P1, $\|r\|^2 \geq 2V_2/m_2$, hence

$$\dot{V}_2 \leq -\frac{2\lambda_{\min}(K)}{m_2} V_2 \quad \implies \quad \|r(t)\| \leq \sqrt{\frac{m_2}{m_1}} \|r(0)\| e^{-\frac{\lambda_{\min}(K)}{m_2} t}$$

$r \rightarrow 0$ exponentially $\Rightarrow e, \dot{e} \rightarrow 0$ exponentially. **Price:** exact knowledge of M, C, G, F .

PD alone with unknown dynamics: ultimate boundedness

Take $\tau = -Kr$ (no model at all):

$M\dot{r} = -Kr - Cr - h(x)$, hence $\dot{V}_2 = -r^TKr - r^Th(x)$.

Bound the cross term by Young's inequality

($\Lambda_1 = \Lambda_1^T > 0$):

$$-r^Th \leq r^T\Lambda_1r + h^T\Lambda_1^{-1}h \leq r^T\Lambda_1r + \bar{\eta}_1$$

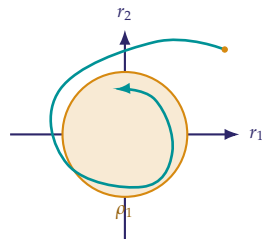
with $\bar{\eta}_1 := \sup_t h^T(x)\Lambda_1^{-1}h(x)$.

UUB result

$$\dot{V}_2 \leq -r^T(K - \Lambda_1)r + \bar{\eta}_1$$

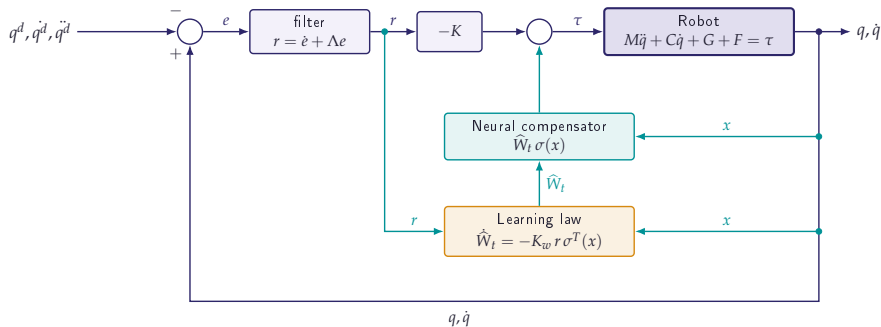
If $K > \Lambda_1$ then $\dot{V}_2 < 0$ whenever $\|r\|_{(K-\Lambda_1)}^2 > \bar{\eta}_1$, so

$$\limsup_{t \rightarrow \infty} \|r(t)\| \leq \sqrt{\frac{\bar{\eta}_1}{\lambda_{\min}(K - \Lambda_1)}} =: \rho_1.$$



The error does *not* vanish: it enters a ball whose radius is set by the *entire* unmodelled dynamics h . Raising K shrinks ρ_1 — at the price of noise amplification, actuator saturation and chatter.

Tracking: PD + neural compensator architecture



Control law

$$\tau = \underbrace{-K r}_{\text{PD: } -K\dot{e} - K\Lambda e} + \underbrace{\hat{W}_t \sigma(x)}_{\text{adaptive feedforward } \approx h(x)}$$

The PD part is *unchanged*: the network is a plug-in feedforward block that can be switched off safely.

Tracking: closed-loop error equation with the compensator

Insert $\tau = -Kr + \hat{W}_t\sigma(x)$ into $M\dot{r} = -Cr + \tau - h(x)$ and use $h = W^*\sigma + \epsilon$:

$$\begin{aligned} M\dot{r} &= -Cr - Kr + \hat{W}_t\sigma(x) - W^*\sigma(x) - \epsilon \\ &= -Cr - Kr + \tilde{W}_t\sigma(x) - \epsilon, \quad \tilde{W}_t := \hat{W}_t - W^*. \end{aligned}$$

Two coupled error systems

$$\begin{cases} M\dot{r} = -(K + C)r + \tilde{W}_t\sigma(x) - \epsilon & \text{(tracking)} \\ \dot{\tilde{W}}_t = \dot{\hat{W}}_t & \text{(parameter)} \end{cases}$$

W^* is constant, so the weight error obeys the learning law directly.

What we must achieve

The term $\tilde{W}_t\sigma(x)$ is *not* sign-definite: it can inject energy. The learning law must be chosen so that it is **exactly cancelled** in $\dot{V}$, leaving only the (small) approximation error ϵ .

The improvement over pure PD: the disturbance dropped from $h(x)$ (*large*) to $\epsilon(x)$ (*small*).

Candidate

$$V_3 = \underbrace{\frac{1}{2}r^T M(q)r}_{\text{tracking}} + \underbrace{\frac{1}{2}\text{tr}(\tilde{W}_t^T K_w^{-1} \tilde{W}_t)}_{\text{parameter estimation}}, \quad K_w = K_w^T > 0.$$

$$\begin{aligned}\dot{V}_3 &= r^T M\dot{r} + \frac{1}{2}r^T \dot{M}r + \text{tr}(\tilde{W}_t^T K_w^{-1} \dot{\tilde{W}}_t) \\ &= -r^T Kr + \underbrace{\frac{1}{2}r^T [\dot{M} - 2C]r}_{=0 \text{ (P2)}} + r^T \tilde{W}_t \sigma(x) - r^T \epsilon + \text{tr}(\tilde{W}_t^T K_w^{-1} \dot{\tilde{W}}_t)\end{aligned}$$

The trace identity that makes it work

$r^T \tilde{W}_t \sigma$ is a scalar, hence $r^T \tilde{W}_t \sigma = \text{tr}(r^T \tilde{W}_t \sigma) = \text{tr}(\tilde{W}_t^T r \sigma^T)$. Therefore

$$\dot{V}_3 = -r^T Kr - r^T \epsilon + \text{tr}\left[\tilde{W}_t^T \left(K_w^{-1} \dot{\tilde{W}}_t + r \sigma^T(x)\right)\right].$$

Tracking: the learning law and the main theorem

Choosing the **gradient-like adaptation**

$$\boxed{\dot{\hat{W}}_t = -K_w r \sigma^T(x)} \implies \text{tr}[\tilde{W}_t^T (K_w^{-1} \dot{\hat{W}}_t + r \sigma^T)] = 0.$$

The remaining term is bounded as before: $-r^T \epsilon \leq r^T \Lambda r + \epsilon^T \Lambda^{-1} \epsilon \leq r^T \Lambda r + \bar{\eta}$.

Theorem (stability of PD + neural compensation)

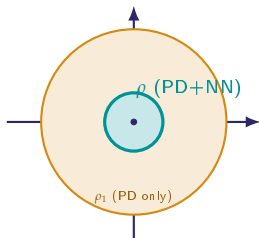
Consider the robot with $\tau = -Kr + \hat{W}_t \sigma(x)$ and $\dot{\hat{W}}_t = -K_w r \sigma^T(x)$. If $K > \Lambda > 0$, then r and $\hat{W}_t$ are *bounded* and the tracking error is *uniformly ultimately bounded*:

$$\dot{V}_3 \leq -r^T (K - \Lambda) r + \bar{\eta}, \quad \limsup_{t \rightarrow \infty} \|r(t)\| \leq \sqrt{\frac{\bar{\eta}}{\lambda_{\min}(K - \Lambda)}} =: \rho,$$

with $\bar{\eta} = \sup_t \epsilon^T \Lambda^{-1} \epsilon$; consequently $\limsup \|e(t)\| \leq \rho / \lambda_{\min}(\Lambda)$.

No persistent excitation is required for *stability*; PE is needed only if $\hat{W}_t \rightarrow W^*$ is desired.

Why learning helps — and how to make it robust



$$\frac{\rho}{\rho_1} \sim \sqrt{\frac{\sup \|\epsilon\|^2}{\sup \|h\|^2}} \ll 1$$

because $h = W^* \sigma + \epsilon$: the network absorbs the *structured* part of h ; only the residual survives, at the same gain K .

Robust modifications (needed in practice)

Pure integral adaptation can drift under noise. Standard fixes:

- ▶ **e -modification:** $\hat{W}_t = -K_w(r\sigma^T + \kappa \|r\| \hat{W}_t)$
- ▶ **σ -modification:** $\hat{W}_t = -K_w(r\sigma^T + \kappa \hat{W}_t)$
- ▶ **Dead-zone:** adapt only when $\|r\| > \rho$
- ▶ **Projection:** keep $\hat{W}_t$ in $\{\|\hat{W}\|_F \leq W_{\max}\}$

Each adds $-\kappa \text{tr}(\tilde{W}^T \hat{W}) \leq -\frac{\kappa}{2} \|\tilde{W}\|_F^2 + \frac{\kappa}{2} \|W^*\|_F^2$ to $\dot{V}_3$, giving UUB in $(r, \tilde{W})$ jointly.

Implementation notes

$\hat{W}_0 = 0$ (start as plain PD) · σ on *normalised* x · discrete update $\hat{W}_{k+1} = \hat{W}_k - T_s K_w r_k \sigma_k^T$, $T_s \lambda_{\max}(K_w) \|\sigma\|^2 < 2$.

What was proved

- 1 **Regulation**, PD + G + F: $\dot{V}_1 = -\dot{q}^T K_d \dot{q} \leq 0$ and LaSalle $\Rightarrow$ *global asymptotic stability*.
- 2 **Regulation**, PD + neural: cross term $\alpha e^T M \dot{q}$ builds $r = \dot{e} + \alpha e$ and gives $\limsup \|e\| \leq 2\bar{\epsilon} / \lambda_{\min}(K_p)$ instead of $(k_g + k_f) / \lambda_{\min}(K_p)$.
- 3 **Tracking**, exact model: $\dot{V}_2 = -r^T K r \Rightarrow$ *exponential convergence*.
- 4 **Tracking**, PD only: UUB with radius $\rho_1 \propto \sup \|h\|$.
- 5 **Tracking**, PD + neural: UUB with radius $\rho \propto \sup \|e\| \ll \rho_1$, no model needed.

The two properties that did all the work

P1 makes $\frac{1}{2} r^T M r$ a Lyapunov function; P2 deletes the Coriolis term in every derivative.

??

Open directions

- ▶ Multilayer σ : inner weights $\Rightarrow$ Taylor residuals
- ▶ Output feedback (no $\dot{q}$): observer + NN
- ▶ RL tuning of K, Λ with a stability certificate

References

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